

FARIA NISHAT KHAN

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RESEARCH FOCUS

Artificial Intelligence (AI) researcher focused on adversarial AI, Trustworthy AI, and AI robustness, with research spanning natural language processing (NLP), large language models (LLMs), and predictive machine learning (ML) for social data. Seeking research opportunities that advance robust and trustworthy AI systems.

EDUCATION

M.S. Computer Science, South Dakota School of Mines and Technology, Rapid City, SD *Expected 2027*
GPA: 4.00/4.00

B.S. Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh *2021*
GPA: 3.74/4.00

RESEARCH EXPERIENCE

Graduate Assistant, South Dakota School of Mines and Technology, Rapid City, SD *Aug 2025 - Present*

- Conduct research under faculty supervision and contribute to academic publications
- Design and run human-subject experiments using VR equipment; affiliated with the VR Lab (Summer 2026 - Present) and the DDIHTS Lab (Aug 2025 - Spring 2026)
- Mentor and support undergraduate and graduate students through office hours; assist faculty with grading

TEACHING EXPERIENCE

Lecturer, Daffodil International University, Dhaka, Bangladesh *Jan 2022 - Jul 2025*

- Developed and delivered undergraduate lectures in Data Structures, Data Mining and Machine Learning, Programming and Problem Solving, and Introduction to Biology and Chemistry for Computation
- Applied gamified learning, flipped classrooms, and student-centric methods to boost engagement
- Supervised 13 final-year students in their Final Year Design Projects; served as Module Leader designing course materials aligned to Course/Program Outcomes
- Affiliated with the DIU NLP and ML Research Lab (Jul 2023 - Jul 2025) and the DIU Health Informatics Research Lab (Jan - Dec 2024)

Teaching Assistant and Teaching Apprentice Fellow, Daffodil International University *Feb 2021 - Dec 2021*

- Supported lecture development and delivery; provided academic guidance to students

Student Prefect (Volunteering TA), Daffodil International University *Jan 2019 - Dec 2020*

- Assisted course teachers with delivery materials and supported students during lab sessions

RESEARCH PROJECTS

MS Thesis: Adversarial Robustness of VR Based Predictive Models (in progress)

- Studying how deep learning models used in VR based prediction systems respond to adversarial perturbation.
- Developing and testing attack methods to evaluate model reliability.

LLM Reliability in Public Health Social Media Content (DDIHTS Lab, awaiting publication)

- Studied how reliably a well-documented GPT model, accessed via Hugging Face, interprets public health content from 2,000 social media posts during the 2024 Philadelphia measles outbreak.
- Ran the model in forced guess mode to systematically probe its hallucination behavior across the dataset.
- Identified two failure modes, default answer concentration and numeric deflation, and used a 5-model ensemble to measure interpretive disagreement across the outbreak lifecycle.

PEER-REVIEWED PUBLICATIONS

- N. Thakur, F. Anderson, **F. N. Khan**, S. N. De Sa, "COVID-19 YouTube Videos in the Late Pandemic: How Faces Detected on Thumbnails and Alignment of Thumbnails with Titles and Descriptions Relate to Engagement?", IEEE CCWC 2026, Las Vegas, NV, pp. 0214- 0223, doi: 10.1109/CCWC67433.2026.11393787.
- N. Thakur, **F. N. Khan**, "Early Reactions to Measles and Chikungunya on TikTok: Which Outbreak Triggered More Fear, Toxicity, and Identity Attack?," IEEE IEMCON 2025, Berkeley, CA, pp. 0329- 0338, doi: 10.1109/IEMCON67450.2025.11381290.

- **F. N. Khan**, S. R. Uddin, Y. A. Shampod, A. Sattar, "The Impact of Corona Pandemic on the E-Commerce Industry Using Data Mining," ICCMC 2021, Erode, India, pp. 1755- 1761, doi: 10.1109/ICCMC51019.2021.9418369.
- M. M. Meherab, N. Faruqui, **F. N. Khan**, T. Islam, S. A. Johan, M. M. Billah, K. S. Rahman, Z. N. M. Z. Mahmud, T. A. Sami, "Dynamic Region-Aware Gradient Suppression (DRAGS): Enhancing Vision Model Robustness by Suppressing Noisy Feature Regions During Training," Proc. Int. Conf. on Intelligent Data Analysis and Applications (IDAA 2025), Dhaka, Bangladesh, 2026, pp. 679- 687, doi: 10.2991/978-94-6239-664-7_47.
- M. U. Hasan, **F. N. Khan**, H. Jahan, Z. Zaman, M. M. Rahman, "Predicting Pancreatic Cancer Survival and Staging: A Machine Learning Approach," IEEE COMPAS 2024, Cox's Bazar, Bangladesh, pp. 1- 5, doi: 10.1109/COMPAS60761.2024.10797176.
- Z. Zaman, T. M. Nova, L. S. Riya, T. Sarker, I. Z. Easha, E. Akter, **F. N. Khan***, "Machine Learning Based Depression Prediction: Comparative Analysis of Models and Stacked Ensemble Approach," ICCIT 2024, Cox's Bazar, Bangladesh, pp. 1416- 1421, doi: 10.1109/ICCIT64611.2024.11021746.
- Z. Zaman, M. H. Meraz, S. Siddike, **F. N. Khan**, M. U. Hasan, M. M. Rahman, "Classifying CCTV Image for Road Accident Analysis Using Deep Learning Techniques," IEEE COMPAS 2024, Cox's Bazar, Bangladesh, pp. 1- 5, doi: 10.1109/COMPAS60761.2024.10796442.

RESEARCH & TECHNICAL SKILLS

Adversarial ML: Universal adversarial perturbations, robustness evaluation, gradient-based attacks (PyTorch implementation)

Deep Learning & Predictive ML: PyTorch, TensorFlow, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, Transformer architectures, scikit-learn, ensemble methods, classification

NLP/LLMs: Topic modeling, sentiment analysis (VADER), transformer-based classification, hallucination analysis, Hugging Face

Systems & Data Tools: Python, Git/GitHub, NumPy, Pandas

LEADERSHIP & PROFESSIONAL SERVICE

- Convener, DIU Girls Computer Programming Club (GCPC), Jul 2023 - Jan 2025; Wing Advisor, Feb - Jul 2023
- Coach, Teaching Apprentice Fellowship, Human Resource Development Institute, DIU, Jan - Dec 2022
- Organizing committee: Data Visionary National Data Analytics Competition (2025), ICPC Asia Regional Dhaka Site (2024), 4th International Robo Tech Olympiad (2024), 2nd Int'l Conf. on Big Data, IoT and ML/BIM (2023), National Girls' Programming Contest (2022)
- CITI Program: Responsible Conduct of Research, NSF Grants (2025); AI-Enhanced Teacher Training Program (2023)